DSANet-ISLES: An Interpretable Multi-Architecture Ensemble of DerNet, SegResNet, and Attention U-Net for Low-Resource Stroke Lesion Segmentation

Abstract. Accurate automated segmentation of ischemic stroke lesions is essential for timely clinical decision-making and treatment planning. However, its practical implementation remains a significant challenge in low-resource clinical settings with limited computational infrastructure. This study proposes DSANet ISLES, an interpretable multi architecture ensemble framework for ischemic stroke lesion segmentation using multi modal MRI data. The proposed method integrates three complementary deep learning architectures, namely DerNet, SegResNet, and Attention U Net, to capture diverse spatial and contextual features. Multi modal information from FLAIR, DWI, and ADC sequences is combined into a unified 3D input representation to enhance lesion characterization. The models are trained independently and their predictions are fused using a weighted ensemble strategy, where optimal weights and threshold values are determined through grid search on the validation set. Additionally, Test Time Augmentation and morphological post processing are applied to improve prediction robustness and reduce false positives. Experiments are conducted on the publicly available ISLES 2022 dataset using a 70%, 15%, and 15% split for training, validation, and testing, respectively. Experimental results on the ISLES 2022 test set show that DSANet-ISLES achieves competitive segmentation performance with a Dice score of 0.8092 and a micro F1-score of 0.8814. The proposed framework demonstrates improved segmentation performance compared to individual models, highlighting the effectiveness of multi architecture fusion in low resource settings. It performs better where imaging quality and lesion presentation differ significantly. Future work will concentrate on combining temporal data and enhancing the model for real-time processing. These improvements will help neuroradiologists manage emergency strokes more effectively in low-resource settings.

Keywords: DenseNet SegResNet Attention U Net Network, Stroke Segmentation, Ensemble Learning

1 Introduction

Ischemic stroke [1] is one of the leading causes of disability and mortality worldwide. Rapid and accurate identification of the affected brain region is essential for effective treatment and prognosis. Automated stroke segmentation can assist clinicians by providing consistent and objective lesion maps, reducing diagnostic time and improving treatment planning. Multi-modal MRI sequences, such as FLAIR, diffusion-weighted imaging (DWI), and apparent diffusion coefficient (ADC), provide complementary information about tissue changes [1,2].

Existing methods have explored both traditional image processing techniques and deep learning approaches. Classical methods often rely on intensity thresholding or region-growing algorithms, which may fail in the presence of noise or subtle lesions. Kuang et al. (2026) [3] proposed LBM-Net, a hybrid CNN-Mamba model, which received an 82% Dice score at ISLES 2022. However, these complex structures represent a large computational cost, and performance reliability may suffer across a wide range of clinical samples. Multi-model ensemble methods provide an essential role in this generalization. Garcia-Salgado et al. [4] present a deep learning-based MRI stroke lesion segmentation method. However, the proposed method have problems of generalization and adaptability. DeepISLES [5] is a powerful multi-modal ensemble-based stroke lesion segmentation method achieving high Dice scores on ISLES 2022. However, its high computational demands and GPU requirements make it challenging for practical use in low resource system. Consequently, there is a need for simple, accurate, and efficient frameworks that can run on limited hardware in real-world healthcare settings.

Addressing these problems is critical for enhancing the reliability and accessibility of automated stroke segmentation. To address those challenges, this paper introduces DSANet-ISLES, a resource-efficient ensemble framework for accurate stroke lesion segmentation. The proposed DSANet-ISLES integrates multi-modal MRI inputs into a unified 3D segmentation framework to enhance diagnostic precision in resource-constrained environments. Three complementary models are trained independently to capture diverse spatial and contextual features of stroke lesions. Their predictions are combined using a weighted ensemble fusion strategy to improve robustness and segmentation accuracy. Advanced preprocessing steps, such as intensity normalization, resampling, and spatial padding, ensure consistency across scans [1]. Additionally, test-time augmentation (TTA) [6] and prediction refinement steps are incorporated to further enhance model generalization. The performance of the framework is evaluated using standard metrics such as Dice [6] and F1-scores [2] to ensure reliable quantitative assessment. The main contributions of this study are summarized as follows:

(i) To develop a lightweight multi-model ensemble framework for improved stroke lesion segmentation,
(ii) To integrate multi-modal MRI data effectively for enhanced feature representation, and
(iii) To design a resource-efficient approach that maintains competitive performance while reducing computational cost.

Related Work

De la Rosa et al. [5] established DeepISLES, an ensemble deep learning model that showed high accuracy in ischemic stroke lesion segmentation and a strong correlation with clinical assessment scores across a variety of real-world datasets. However, the model still faces challenges in accurately detecting small infarcts in regions prone to imaging artifacts or in cases involving complex imaging patterns from chronic lesions. Zelin et. al [7] proposes a W-Net model for stroke lesion segmentation and achieves about 61.76% Dice score with improved boundary accuracy compared to existing methods. However, their model depends on complex design and does not focus on a simple and computationally efficient approach for practical use.

Zhang et al. [8] developed a universal foundation model called Mixture of Modality Experts (MoME) that utilizes specialized expert networks and a hierarchical gating system to automatically segment various brain lesion types across diverse MRI modalities. While this approach demonstrates high accuracy and strong generalization, it primarily relies on 3D volumetric data and may encounter computational constraints when attempting to integrate larger numbers of modality-specific experts or learning from extremely high-resolution context sets. Jazzar and

Douik et. al [9] developed the Res2U++ model using a multi-scale encoder and an ASPP block to achieve a training Dice score of 0.85 on the ISLES 2022 dataset. However, their architecture relies on complex feature extraction that requires high-end computing power, making it difficult to implement in low-resource countries where the necessary hardware is often unavailable.
Pacal et al. [10] achieved a high Dice score of 0.89 on the ISLES 2022 dataset by using gated convolutions and multi-scale attention to effectively capture both small and large stroke lesions. However, their model's reliance on 65.12 million parameters and specialized hardware for fast inference makes it difficult to deploy in low-resource countries where advanced computing power and high-end GPUs are often unavailable. Rahman et al. [11] introduced a multi-channel deep learning model using DenseNet121 and SelfONN to achieve a high Dice score of 87.49% on the ISLES 2022 dataset. However, their approach relies on expensive, high-end GPUs and the fusion of multiple MRI sequences, which are often unavailable in low-resource clinical settings. Relying only on single-modality data in these regions would significantly drop the model's performance, as it lacks the multi-spectral information needed to accurately distinguish stroke lesions from surrounding brain tissue.
Chu et al. [12] introduced StrokeFlow to improve boundary accuracy and the detection of small lesions on the ISLES 2022 dataset. However, the model requires specialized supervision and alignment with ADC maps, which can be difficult to implement in facilities where multi-modal data is inconsistent. Furthermore, the paradigm shift toward continuous field modeling involves complex mathematical computations that may not be sustainable in low-resource settings lacking advanced processing hardware. Existing models frequently achieve high accuracy by employing complex architectures that need large processing resources and costly GPUs. However, these models are challenging to implement in countries with limited resources where advanced technology and comprehensive multi-modal MRI sets are often unavailable. Moreover, these models operate as black-box systems, making their decision-making process difficult to interpret. As a result, there is a growing demand for efficient, single-network solutions that produce consistent results on conventional clinical equipment without the need for expensive infrastructure.

3 Methodology

The proposed method starts from multi-modal data acquisition and preprocessing to ensemble-based segmentation and evaluation. The overall working procedure consists of modality pairing, preprocessing, independent model training, ensemble fusion, post-processing, and quantitative assessment. **Figure 1** shows the step-by-step procedure of segmenting the stroke lesion.

3.1 Data Preprocessing

This study use the ISLES-2022 dataset [1] for experiment. Then normalize the intensity values of all images and apply Z-score normalization [13]. The formula is $X_{norm} = \frac{X-\mu}{\sigma}$ Here, μ is the mean intensity and σ is the standard deviation. To remove excess background noise and fit the GPU memory limits, this study apply spatial padding and random cropping [13]. The study resize all volumes to a fixed dimension of *64*64*64* and this uniform size ensures consistent model input. Furthermore, since medical datasets typically contain a limited amount of data, spatial augmentations are applied to prevent overfitting. Specifically, flips along the X, Y, and Z axes are performed using 'RandFlipd' [14] with a 50% probability, and random 90-degree rotations are executed using 'RandRotate90d' [14] with the same probability. Finally, to enhance memory efficiency and ensure rapid computations on the GPU, all data is converted to the torch.float32 format using the 'CastToTyped' function. **Figure 2** shows the step by step preprocessing stage.

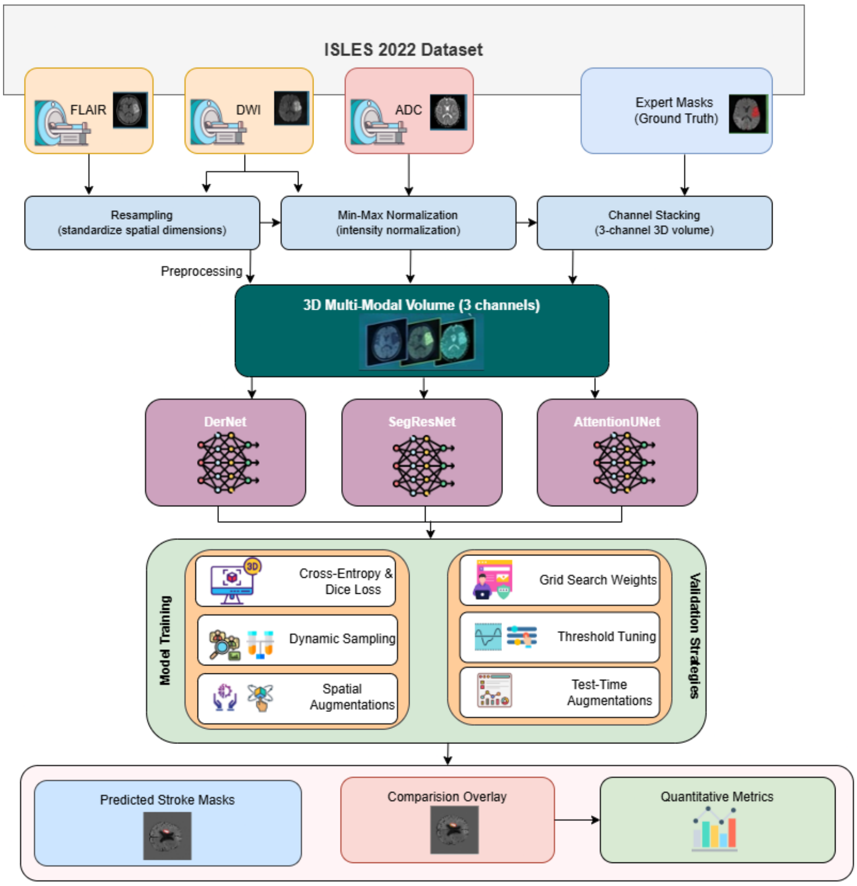


Figure 1: Step-by-step procedure of of stroke lesion segmentation

3.2 Problem Formulation

We aim to segment lesions from medical images. Let an input image be $X \in R^{H*W*D*C}$. Here, H, W, D represent the spatial dimensions. C represents the number of input modalities. The true label mask is $Y \in \{0,1\}^{H*W*D}$. The goal is to learn a mapping function f_θ. This function predicts a binary mask $\hat{Y}$. The equation is $\hat{Y} = f_\theta(X).\theta$. θ represents the trainable model parameters. We optimize θ to minimize the difference between Y and $\hat{Y}$.

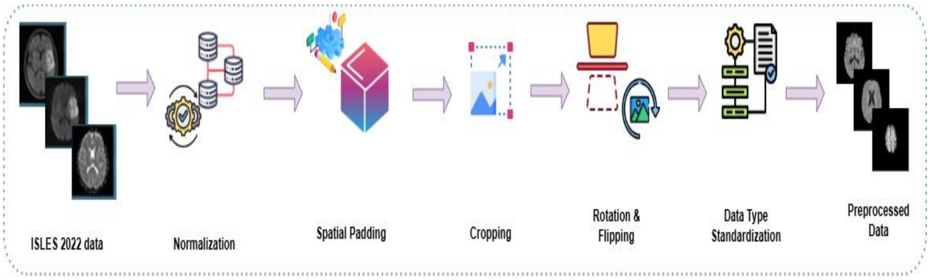


Figure 2: Overview of the 3D MRI data preprocessing pipeline

3.3 Proposed DSANet Framework

We propose the DSANet framework for the stroke lesion segmentation. The name stands for DerNet [15], SegResNet [16], and Attention U-Net [17] Network. DSANet uses a multi-branch architecture and extracts diverse features from the input images. The framework consists of three parallel branches. Each branch generates an independent prediction mask. An ensemble module combines these predictions. This combination improves the final segmentation accuracy. The DSANet consists of three distinct deep learning architectures.

3.3.1 DerNet

This study use a DerNet model which connects each layer to every other layer. This structure ensures maximum information flow. The feature map of the l-th layer is x_l. The operation is $x_l = H_l([\{x_0, x_1, ..., x_{l-1}\}])$. H_l denotes a composite non-linear function. The brackets denote feature concatenation. DenseNet effectively extracts high-level semantic features.

This study set the random seed to 42 for complete reproducibility and normalize the intensity values of all input images. Then this study crop the image volumes to a fixed dimension of *64x64x64* and apply random flips and spatial rotations to the training data. These augmentations prevent the model from overfitting. The study initialize the network using pre-trained weights and fine-tune the model for 80 epochs. Then this study use the AdamW optimizer [18] with a learning rate of *1e-4* and optimize the model using the DiceFocalLoss function. This specific loss function handles target class imbalance effectively. This study also use Automatic Mixed Precision (AMP) during the training phase. AMP speeds up computation and reduces memory usage. The specific hyperparameter configurations used for training the DERNet model are summarized in **Table 1**.

Table 1: Hyperparameter values of the DERNet model.

Parameter	Value
Input ROI Size	64 x 64 x 64
Batch Size	1
Total Epochs	80
Initial Learning Rate	1e-4
Optimizer	AdamW
Loss Function	DiceFocalLoss

Evaluation Metric	Dice Metric (F1 Score)
Train/Val/Test Split	70% / 15% / 15%
Random Seed	42
Augmentation Probability	0.5 (Flip & Rotate)

3.3.2 SegResNet

This study uses the SegResNet model as the second model of the segmentation framework. SegResNet utilizes residual connections and a residual block computes $y = F(x, Wi) + x$. Here, x is the input and y is the output. $F(x, W_i)$ represents the residual mapping. These connections solve the vanishing gradient problem. SegResNet captures multi-scale contextual information. The network is initialized with specific architectural parameters, including an initial filter size of 32 and a dropout probability of 0.2. The global random seed is set to 42 to ensure the reproducibility of the experiments. During the data preprocessing phase, the intensity of all input images is normalized, and the volumes are padded or cropped to a uniform spatial dimension of 64x64x64. The training data is augmented using random spatial crops, random flips, and 90-degree rotations with a 50% probability to enhance model generalization.

The study trains the SegResNet model for 100 epochs using the AdamW optimizer with an initial learning rate of 1e-4 and a cosine annealing learning rate scheduler. To optimize memory usage while maintaining training stability, a gradient accumulation strategy is utilized. The physical batch size is set to 1, but gradients are accumulated over 4 steps, effectively simulating a larger batch size of 4. The model weights are optimized using the DiceFocalLoss function. The training process is accelerated by applying Automatic Mixed Precision (AMP) [19] with a gradient scaler. The model's performance is evaluated on the validation set at the end of each epoch using a sliding window inference technique. Finally, to maximize the evaluation accuracy on the test set, Test-Time Augmentation (TTA) [19] is applied. This technique processes the original image alongside its X-axis and Y-axis flipped versions, averaging the three predictions to produce a highly precise final segmentation mask. The specific hyperparameter configurations used for training the SegResNet model are summarized in **Table 2**.

Table 2: Hyperparameter values of the SegResNet model.

Parameter	Value
Input ROI Size	64 x 64 x 64
Physical Batch Size	1
Gradient Accumulation Steps	4 (Simulated Batch Size = 4)
Total Epochs	100
Initial Learning Rate	1e-4
Optimizer	AdamW (Weight Decay = 1e-4)
LR Scheduler	Cosine Annealing
Loss Function	DiceFocalLoss (γ = 2.0)
Evaluation Metric	Dice Metric (F1 Score)
Train/Val/Test Split	70% / 15% / 15%
Random Seed	42
Augmentation Probability	0.5 (Flip & Rotate)
Dropout Probability	0.2

3.3.3 Attention U-Net

The study uses the Attention U-Net model as the third model of the segmentation framework. The model incorporates attention gates to highlight salient features. These gates effectively suppress irrelevant background regions. Attention gates highlight salient features. They suppress irrelevant background regions. The attention coefficient is α_i. We calculate it as $\alpha_i = \sigma(W^T \tanh(W_x x_i + W_g g_i + b_1) + b_2)$. Here, x_i is the input feature, g_i is the gating signal and σ is the sigmoid function. The random seed is set to 42 to guarantee reproducible outcomes. During preprocessing, the intensity of all input images is normalized. The image volumes are padded or cropped to a uniform spatial dimension of 64x64x64. The training dataset undergoes several data augmentation techniques to prevent overfitting. These techniques include random spatial crops, random flips, and 90-degree rotations with a 50% probability.

The Attention U-Net is trained for exactly 100 epochs. The training process uses a batch size of 1. The model parameters are optimized using the AdamW optimizer with a weight decay of 1e-5. The initial learning rate is set to 1e-4 and is regulated by a cosine annealing scheduler. The network learns to segment lesions using the DiceFocalLoss function. The training loop utilizes Automatic Mixed Precision (AMP) with a gradient scaler to maximize convergence and minimize memory consumption. The model is evaluated on the validation set at the end of each epoch using a sliding window inference approach. The best-performing model weights, measured by the highest Dice Similarity Coefficient (F1 Score), are saved for final testing. The specific hyperparameter configurations used for training the Attention U-Net model are summarized in **Table 3.**

Table 3: Hyperparameter values of the Attention U-Net model.

Parameter	Value
Input ROI Size	64 x 64 x 64
Batch Size	1
Total Epochs	100
Initial Learning Rate	1e-4
Optimizer	AdamW (Weight Decay = 1e-5)
LR Scheduler	Cosine Annealing
Loss Function	DiceFocalLoss (γ = 2.0)
Evaluation Metric	Dice Metric (F1 Score)
Train/Val/Test Split	70% / 15% / 15%
Random Seed	42
Augmentation Probability	0.5 (Flip & Rotate)

3.4 Ensemble Strategy

This study proposes an advanced ensemble pipeline to maximize final prediction accuracy. First, TTA is applied to the test set. Each model processes the original image, an X-axis flipped version, and a Y-axis flipped version. The pipeline averages these three predictions. Next, an automated grid-search runs on the validation set to find the best combination weights for the three models. This search gives the highest weight priority to the DERNet model. Another grid-search finds the optimal threshold value. This step converts the weighted probability map into a final binary mask. Finally, a morphological post-processing step removes small, disconnected noise blobs under 30 pixels. This clean-up eliminates false positives and heavily refines the prediction. Here, this study combine the outputs of the three branches. Let the three individual predictions be P_1, P_2, P_3. We use a weighted averaging strategy. The final prediction is P_{final}. We calculate it as $P_{final} = w_1 P_1 + w_2 P_2 + w_3 P_3$. Here, w_1, w_2, w_3 are learned weights and $w_1 + w_2 + w_3 = 1$. We apply a threshold T to P_{final}. This step creates the final binary mask. Algorithm 1 shows how the full procedure works.

Algorithm 1: Proposed Ensemble Segmentation Pipeline

Input: *Validation dataset* $D_{val} = \{(X_i, Y_i)\}$, *Test images* $D_{test} = \{X_j\}$, *Trained models* $M = \{M_{DER}, M_{Seg}, M_{Att}\}$.

Output: *Final segmented binary masks* $\hat{Y}_{test}$ *for the test set.*

Phase 1: Hyperparameter Optimization via Grid-Search (Validation Set)

1. ***Initialize*** *maximum Dice score* $S\_max \leftarrow 0$, *optimal weights* $W^* \leftarrow \{0, 0, 0\}$, *optimal threshold* $T^* \leftarrow 0$.
2. ***For*** *each candidate weight combination (w1, w2, w3) such that* $\sum wk = 1$ *and* $w1 \geq 0.4$:
    3. ***For*** *each candidate threshold* $T \in \{0.35, 0.40, 0.45, 0.50\}$:
        4. ***For*** *each validation sample* $(X_i, Y_i) \in D_{val}$:
            5. *Compute Test-Time Augmentation (TTA) predictions* $P_{k,i}$ *for each model* $M_k \in M$
            6. *Compute weighted probability map:* $P_{ens,i} = w1 \cdot P_{DER,i} + w2 \cdot P_{Seg,i} + w3 \cdot P_{Att,i}$
            7. *Generate binary mask:* $B_i = 1$ *if* $P_{ens,i} > T$ *else 0*
        8. *End* ***For***
        9. *Calculate overall validation Dice score* S_{val} *using all predictions* $\{B_i\}$ *and ground truths* $\{Y_i\}$
        10. ***If*** $S_{val} > S_{max}$ *then:*
            11. $S_{max} \leftarrow S_{val}$
            12. $W^* \leftarrow \{w1, w2, w3\}$, $T^* \leftarrow T$
        13. *End* ***If***
    14. *End* ***For***
15. *End* ***For***

Phase 2: Final Inference and Morphological Post-Processing (Test Set)

16. ***For*** *each test sample* $X_j \in D\_test$:
    17. ***For*** *each model* $M_k \in M$:
        18. *Compute base prediction:* $p_{orig} = M_k(X_j)$
        19. *Compute X-axis flipped prediction:* $p_x = FlipX_{inv}(M_k(FlipX(X_j)))$
        20. *Compute Y-axis flipped prediction:* $p_y = FlipY_{inv}(M_k(FlipY(X_j)))$
        21. *Average TTA prediction:* $P_{k,j} = (p_{orig} + p_x + p_y) / 3$
    22. *End* ***For***
    23. *Compute optimal weighted map:* $P^*_{ens,j} = w1^* \cdot P_{DER,j} + w2^* \cdot P_{Seg,j} + w3^* \cdot P_{Att,j}$
    24. *Apply optimal threshold:* $B^*_j = 1$ *if* $P^*_{ens,j} > T^*$ *else 0*
    25. *Apply morphological filtering:* $\hat{Y}_j = RemoveSmallObjects(B^*_j, min\_size=30)$

*26. End **For***
*27. **Return** set of final predicted masks {$\hat{Y}_j$}*

Several deep learning models were tested to find the most effective way to segment stroke lesions. These models included SegResNet, Attention U-Net, DERNet, and DynUNet 20]. After the initial runs, DERNet, SegResNet, and Attention U-Net were selected due to their high accuracy and ability to capture lesion edges. These three models were combined into an ensemble to correct individual errors and reduce false results. Other models, such as DynUNet, were excluded because of their high complexity and lower performance on the dataset present in **Table 4**.

Table 4: Performance comparison of different deep learning models

Model Name	Best Validation Dice	Final Test Dice
DERNet	0.8215	0.8136
SegResNet	0.7801	0.7819
Attention U-Net	0.7789	0.7274
DynUNet	0.6819	0.5796
Modified DynUNet	0.6164	0.5271

The ensemble model is used to combine the unique architectural strengths of multiple models to achieve a more stable and balanced prediction. By merging the outputs of different networks, individual errors and biases are reduced, leading to more reliable segmentation across various lesion shapes and sizes.

3.5 Training Strategy & Low-Resource Learning Strategy

This study employs a robust training strategy. The loss function combines Dice Loss [21] and Cross-Entropy Loss [21]. The total loss is L_{total} and $L_{total} = \lambda_1 L_{Dice} + \lambda_2 L_{CE}$. Here, λ_1 and λ_2 are balancing weights. This study use the AdamW optimizer. The initial learning rate is $1*10^{-4}$ and train the models for 100 epochs. This study also use gradient accumulation to simulate a larger batch size. A Cosine Annealing scheduler controls the learning rate. It smoothly decreases the learning rate during training. The models use the DiceFocalLoss function for error calculation. Medical image segmentation often has a huge class imbalance between the lesion and the background. The DiceFocalLoss effectively handles this imbalance problem. The study uses Automatic Mixed Precision (AMP) and a gradient scaler. These tools speed up the training process. They also reduce GPU memory consumption. Hardware limitations prevent the use of large batch sizes. Therefore, the study applies a gradient accumulation technique. This technique ensures stable training by simulating a larger batch size. A sliding window inference is run on the validation set at the end of each epoch. The model weights with the highest F1-score are saved for final testing.

Medical image datasets generally contain a small amount of data. This study adopts a low-resource learning strategy to overcome this limitation. Several data augmentation techniques are applied during training. These techniques prevent the models from memorizing the training data. First, every input image is preprocessed to a uniform 64x64x64 dimension. Next, random spatial crops are used during training. The study also applies random flips and 90-degree spatial rotations with a 50% probability. These augmentations help the models recognize new patterns. They significantly increase the generalization power on unseen data. The study utilizes TTA during the testing phase. TTA creates multiple augmented versions of a single test image. For example, it generates an original version, an X-axis flipped version, and a Y-axis flipped version. The network processes all three versions independently. The pipeline then averages these three predictions. This specific method drastically boosts the final segmentation accuracy without requiring any extra training data.

3.6 Evaluation Protocol and Implementation Details

We split the dataset deterministically. We use a 70-15-15 split. Seventy percent of the data is for training. Fifteen percent is for validation. Fifteen percent is for testing. We set the random seed to 42. This seed guarantees reproducible results. We save the model with the best validation score. We evaluate this best model on the unseen test set. We measure performance using three primary metrics. We use the Dice Similarity Coefficient (DSC). DSC measures spatial overlap. The formula is $DSC = \frac{2\times|Y\cap\hat{Y}|}{|y|+|\hat{Y}|}$.Here, Y is the ground truth and $\hat{Y}$is the prediction. We also calculate Precision, recall and f1 score.

We implement the framework using PyTorch. We use the MONAI library for medical image processing. We run the experiments on a single NVIDIA GPU. We use mixed-precision training. This technique speeds up the training process. It reduces memory consumption.

4 Results

This section presents a comprehensive evaluation of the developed ensemble model on the ISLES-2022 dataset. First, the quantitative performance is assessed using standard medical image segmentation metrics to demonstrate the model's spatial accuracy and reliability. Following this, a qualitative analysis is provided to visually inspect the boundary delineation capabilities and network interpretability. Finally, the clinical relevance, architectural contributions, and the broader implications of the findings are discussed.

4.1. Quantitative Performance Evaluation

The performance of the DSANet-ISLES was evaluated on the test set. The final model utilized an optimized weighted ensemble strategy and an optimal cutoff threshold of 0.5, determined through an automated grid search on the validation set. The quantitative findings are presented using two complementing averaging methods: macro average (subject-wise mean) and micro average (global voxel-wise aggregate). The Mean Subject-wise Dice Score evaluates reliability across patients, whereas the global voxel-wise metrics give insights into overall pixel-level accuracy. The detailed performance metrics are presented in **Table 5**.

Table 5: Quantitative Segmentation Performance on the Test Set

Metric (Averaging Method)	Mean Score (0 to 1)
Dice Similarity Coefficient (Macro)	0.8092
Intersection over Union (IoU - Macro)	0.6991

Precision (PPV - Macro)	0.8848
Recall (Sensitivity - Macro)	0.7842

To further analyze the pixel-level performance, a global confusion matrix was calculated. This matrix covers all test subjects and includes over 32.6 million voxels. The model achieved 106,024 true positives, 19,093 false positives, 9,430 false negatives, and 32,507,209 true negatives shown in Figure 3. The micro F1 Score is 0.8814, which indicates a high overlap between the predicted stroke areas and the actual ground truth. Furthermore, the model achieved a micro Recall of 0.9183 which significantly reducing the risk of missing a critical lesion. Additionally, the micro Precision score of 0.8474 shows that when the model predicts a stroke. The large number of True Negatives compared to the very low False Positives confirms that the model effectively handles the severe class imbalance and successfully ignores healthy brain tissue.

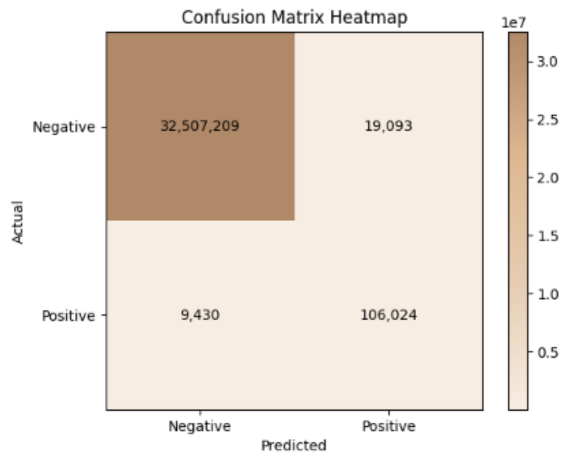
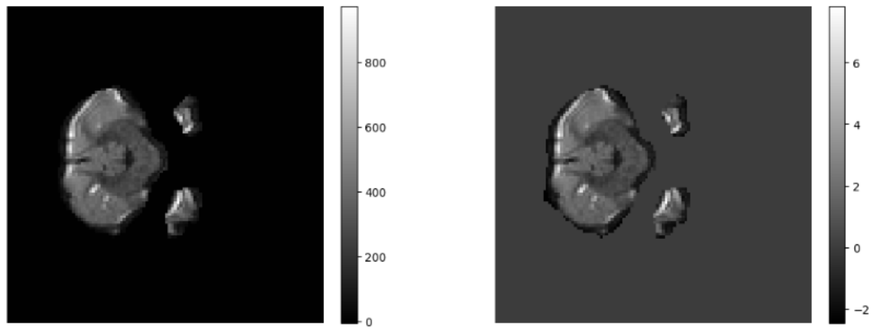


Figure 3: Normalized voxel-wise confusion matrix across the entire ISLES-2022 test set. The high true negative rate shows the effectiveness in suppressing healthy tissue background.

4.2. Qualitative Analysis and Boundary Analysis

A detailed qualitative analysis was also performed. This visual check evaluated the model's ability to outline lesion boundaries accurately. The data preprocessing pipeline was tested. The effect of intensity normalization on a raw DWI input was visualized. **Figure 4** compares the raw DWI image with its Z-score normalized version. The normalized image shows better contrast. This ensures that intensity changes do not reduce the segmentation accuracy. An advanced 6-panel boundary analysis grid was created. This grid compares the model's predictions with the expert ground truth. An example of an axial slice with stroke lesions is shown in **Figure 5**. The analysis grid shows two main points. First, panel (d) shows a tight overlap between predicted and ground truth boundaries. Second, panel (f) shows the Layer-CAM visualization. It confirms that the model focuses specifically on the stroke areas. It ignores other healthy brain parts. This proves the strong detection ability of the proposed method. **Figure 6** shows the qualitative segmentation results, where the red contours represent the predicted stroke lesions. It demonstrates the model's ability to accurately identify multifocal and irregular ischemic regions across different brain slices.



(a) Raw DWI (b) Normalized DWI

Figure 4: Verification of the data pre-processing pipeline. Panel (a) shows the original raw DWI input with a high intensity range. Panel (b) displays the corresponding Z-score normalized input with a standardized intensity distribution.

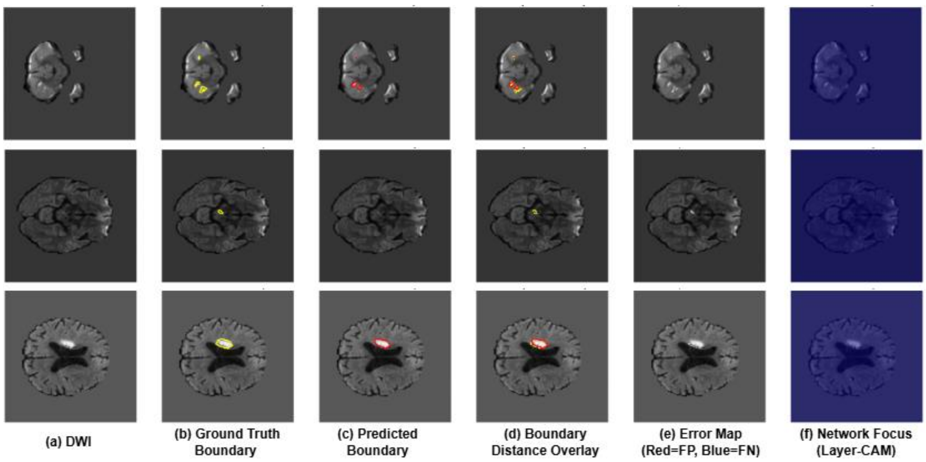


Figure 5: Advanced Boundary and Focus Analysis on an Axial DWI slice. (a) Input DWI; (b) Ground truth lesion boundary (yellow contour); (c) Predicted lesion boundary (red contour); (d) Boundary Distance Overlay comparing overlap; (e) Error Map highlighting False Positives (Red) and False Negatives (Blue); (f) Network Focus visualized using Layer-CAM.

Figure 5 (e) shows the spatial error distribution, reflecting 19,093 false positive and 9,430 false negative voxels across the test set. The higher count of false positives indicates a slight tendency of the model to over-segment healthy areas around the lesion boundaries. However, the visually minimal false negatives align with the high Recall 0.9183, showing that the model rarely misses critical stroke areas, thereby ensuring high clinical safety.

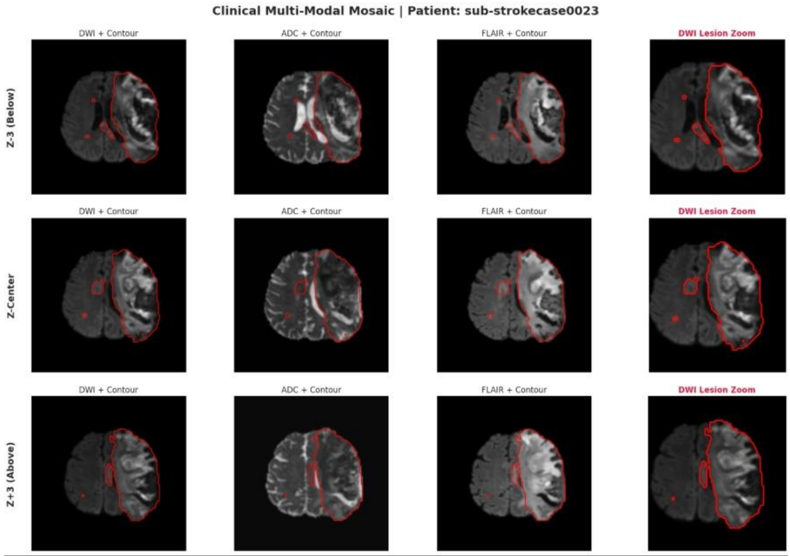


Figure 6: Qualitative segmentation results on DWI slices from the ISLES 2022 dataset.

4.3 Discussion

The results show that the proposed DSANet-ISLES model performs very well on the ISLES-2022 dataset. A Macro Dice Score of 0.8092 proves accurate lesion detection across different patients. The difference between the Macro Average (0.8092) and the Micro Average (~0.88) is important. The Micro Average gives a higher score because it favors larger lesions with more pixels. However, the Macro Average treats every patient equally, regardless of the lesion size. The close gap between these two scores is a positive sign. It means the model performs well on both large and small lesions. In medical clinics, a high Precision (0.8848) is very important. A bad model might predict healthy brain tissue as a stroke. This false alarm causes patient stress and costs extra money. The current model has a very low False Positive count (19,093 compared to 32.5 million True Negatives). This greatly reduces clinical errors. The automated weight optimization also gave interesting results. It assigned a high weight (0.8) to DERNet. AttentionUnet received a small weight (0.2), and SegResNet received zero (0.0). This shows that DERNet is highly effective for stroke segmentation. AttentionUnet adds some helpful features, but SegResNet is not needed in this setup. In the future, the model can be improved further. Small lesions with blurry boundaries are still hard to detect accurately. Future work will add edge-aware loss functions to solve this problem.

5 Conclusion

The study presents a practical approach for ischemic stroke lesion segmentation that balances performance with computational feasibility in limited resource settings. By combining different model structures with complementary strengths and refining their outputs through validation driven weighting and simple enhancement steps, the framework shows that reliable segmentation can be achieved without relying on heavy or complex systems. The design also keeps the process interpretable and modular, which makes it easier to adapt or extend in future work. This proposed method supports the development of more accessible medical imaging solutions and opens the path for further improvements through better data use, lightweight optimization, and broader clinical validation across diverse environments.

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References

1. Hernandez Petzsche, M.R., De La Rosa, E., Hanning, U., Wiest, R., Valenzuela, W., Reyes, M., et al.: ISLES 2022: A multi-center magnetic resonance imaging stroke lesion segmentation dataset. Scientific Data 9(1), 762 (2022)
2. Zhao, R., Li, Z., Xue, Z., Ohtsuki, T., Gui, G.: A novel approach based on lightweight deep neural network for network intrusion detection. In: IEEE Wireless Communications and Networking Conference (WCNC), pp. 1–6. IEEE, New York (2021)
3. Kuang, Z., Yan, X., Yu, J., Sun, D., Zhao, J., Sun, L.: LBMNet: A hybrid multi-scale CNN–Mamba framework for enhanced 3D stroke lesion segmentation in MRI. In: Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), LNCS. Springer, Heidelberg (2026)
4. Garcia-Salgado, B.P., Almaraz-Damian, J.A., Cervantes-Chavarria, O., Ponomaryov, V., Reyes-Reyes, R., Cruz-Ramos, C., Sadovnychiy, S.: Enhanced ischemic stroke lesion segmentation in MRI using attention U-Net with generalized dice focal loss. Applied Sciences 14(18), 8183 (2024)
5. De La Rosa, E., Reyes, M., Liew, S.L., Hutton, A., Wiest, R., Kaesmacher, J., et al.: DeepISLES: a clinically validated ischemic stroke segmentation model from the ISLES'22 challenge. Nature Communications 16(1), 7357 (2025)
6. Hoar, David, Peter Q. Lee, Alessandro Guida, Steven Patterson, Chris V. Bowen, Jennifer Merrimen, Cheng Wang, Ricardo Rendon, Steven D. Beyea, and Sharon E. Clarke. "Combined transfer learning and test-time augmentation improves convolutional neural network-based semantic segmentation of prostate cancer from multi-parametric MR images." *Computer Methods and Programs in Biomedicine* 210 (2021): 106375.
7. Wu, Zelin, Xueying Zhang, Fenglian Li, Suzhe Wang, Lixia Huang, and Jiaying Li. "W-Net: A boundary-enhanced segmentation network for stroke lesions." *Expert Systems with Applications* 230 (2023): 120637.
8. Zhang, Xinru, Ni Ou, Berke Doga Basaran, Marco Visentin, Mengyun Qiao, Renyang Gu, Paul M. Matthews, Yaou Liu, Chuyang Ye, and Wenjia Bai. "A foundation model for lesion segmentation on brain mri with mixture of modality experts." *IEEE Transactions on Medical Imaging* (2025).
9. Jazzar, Nesrine, and Ali Douik. "Res2U++: Deep learning model for segmentation of ischemic stroke lesions." *Biomedical Signal Processing and Control* 102 (2025): 107269.

10. Pacal, Ishak, Ali Algarni, and Ismail Kunduracioglu. "Adaptive and efficient deep learning model for automated ischemic stroke lesion segmentation." *Biomedical Signal Processing and Control* 118 (2026): 109658.
11. Rahman, Ashiqur, Muhammad EH Chowdhury, Md Sharjis Ibne Wadud, Rusab Sarmun, Adam Mushtak, Sohaib Bassam Zoghoul, and Israa Al-Hashimi. "Deep learning-driven segmentation of ischemic stroke lesions using multi-channel MRI." *Biomedical Signal Processing and Control* 105 (2025): 107676.
12. Chu, Liuxi, Ying Wang, Zhijin Li, Xiaotong Liu, Shui Tian, Hongqiang Xie, and Yalin Zhang. "Modeling Ischemic Stroke Pathological Dynamics via Continuous Fields and Vector Flow." *npj Digital Medicine* 9, no. 1 (2026): 50.
13. Krotov, Aleksei, Reza Sharif Razavian, Mohsen Sadeghi, and Dagmar Sternad. "Time-warping analysis for biological signals: methodology and application." *Scientific Reports* 15, no. 1 (2025): 11718.
14. Li, Tong, Renguang Zuo, Yihui Xiong, and Yong Peng. "Random-drop data augmentation of deep convolutional neural network for mineral prospectivity mapping." *Natural Resources Research* 30, no. 1 (2021): 27-38.
15. Wang, Dingyu, Shaocheng Jia, Xin Pei, Chunyang Han, Danya Yao, and Dezhi Liu. "Dernet: Driver emotion recognition using onboard camera." *IEEE Intelligent Transportation Systems Magazine* 16, no. 2 (2023): 117-132.
16. Cariola, Annachiara, Elena Sibilano, Andrea Guerriero, Vitoantonio Bevilacqua, and Antonio Brunetti. "Deep learning strategies for semantic segmentation of pediatric brain tumors in multiparametric MRI." *Scientific reports* 15, no. 1 (2025): 22595.
17. Xu, Qing, Zhicheng Ma, and Wenting Duan. "DCSAU-Net: A deeper and more compact split-attention U-Net for medical image segmentation." *Computers in biology and medicine* 154 (2023): 106626.
18. Zhou, Pan, Xingyu Xie, Zhouchen Lin, and Shuicheng Yan. "Towards understanding convergence and generalization of AdamW." *IEEE transactions on pattern analysis and machine intelligence* 46, no. 9 (2024): 6486-6493.
19. Gupta, Akansha, and Arjun Singh Rawat. "Comparative Analysis of Deep Learning Models for Lesion-Wise Segmentation of Diabetic Retinopathy." *IEEE MultiMedia* (2026).
20. Deng, Yu, Yiyang Xu, Linglong Qian, Charlène Mauger, Anastasia Nasopoulou, Steven Williams, Michelle Williams et al. "MorphiNet: A Graph Subdivision Network for Adaptive Bi-ventricle Surface Reconstruction." *IEEE Transactions on Medical Imaging* (2026).
21. Yeung, Michael, Evis Sala, Carola-Bibiane Schönlieb, and Leonardo Rundo. "Unified focal loss: Generalising dice and cross entropy-based losses to handle class imbalanced medical image segmentation." *Computerized Medical Imaging and Graphics* 95 (2022): 102026.